

The Essence of Being Necessary

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In his 1994 paper *Essence and Modality* Kit Fine argued against the view that essence should be understood in terms of *de re* modality. He suggested the conversed account: the metaphysical necessity should be understood in terms of essence, of what is true in virtue of the nature of what. The broad tendency to explain modality in terms of essence was followed by philosophers such as Hale and Lowe, albeit some differences in details. I will classify all such accounts as *essentialism* on modality.

The obvious challenge for essentialists would be to provide an account on essence that does not rely on modal notions, and at the same time to derive the fact that p being necessarily true from the fact that p being essentially true. Skeptics argue that this was never done. The arguments provided for the latter implicitly assumes some modal quality on the essentiality of p , and that is to fail to satisfy the former requirement. And if the essentiality of p has no modal quality at all, there seems to be a gap between something being essential and being necessary, so that the derivation from essentiality to necessity is impossible to achieve.

In this paper, I will argue that essentialists can reply to the skeptic challenge without compromising any of their core thesis. The skeptics overlook one remaining possibility: that the gap is filled by the essentialist quality on the notion of something being metaphysically necessary. More specifically, I suggest that both requirements can be satisfied by assuming that it is true in virtue of the nature of being necessary that *if something is essentially true, then it is necessarily so*. After some preliminary clarifications in section 1, I will argue in section 2 that any attempts on non-transmissive explanation of why the necessary fact is necessary, including the essentialist view, would ultimately become the account on the essence of being necessary. I will reformulate the essentialist claim in terms of the essence of being necessary in section 3. In section 4, I will clarify why such an account does not cause any kinds of trouble on circularity. Note that I am not trying to argue for the essentialist answer for the problem of the source of necessity, against other alternatives.

1 The Source of Necessity

For any necessary fact p , we can ask why is p necessary. The question is different from the question of why does p obtain. For example, we can answer the question of why is it the case that $3^2 = 9$ or snow is white by the truth of $3^2 = 9$. But this cannot be the answer for the question of why is it necessarily the case that $3^2 = 9$ or snow is white.

Similarly, the question of the source of *particular* necessity should be distinguished from the question of the *general* source of necessity. It might be possible to address the source of the particular necessity of the previous fact by the necessity of $3^2 = 9$. This however does not guide us at all about the general source of necessity, since the necessity of $3^2 = 9$ is left unexplained. The general source of necessity should be able to provide, mediately or immediately, the source of all necessity.

One difficulty answering the question is noted by Blackburn (1993). He argued that the general source of necessity cannot be explained by some true propositions. Suppose we want to answer the question by pointing to some true propositions, that is, to claim that $\Box p$ because q . It would be obvious that q is either necessarily true or contingently true. If it is necessarily true, then it cannot be the final source of necessity, for it leaves the necessity of q unexplained. But it also cannot be the case that q is contingently true, for “... then there is strong pressure to feel that the original necessity has not been explained or identified, so much as undermined” (Blackburn 1993, p.53)

Blackburn wishes to further conclude that therefore any realist conception on necessity must fail. This advance might be too hasty. One can insist that necessity is metaphysically primitive, therefore not explainable by other notions. It also might be suggested that the source of necessity need not be truth-bearers such as propositions. Not every possibility is closed by the proposed dilemma.

However, it is also argued that to explain necessity via contingently or necessarily true proposition is indeed possible. Hale (2002) pointed out that the denial of contingently true propositions explaining necessity presupposes that necessity is never contingent, which some philosophers, notably conventionalists would disagree. Still, Hale himself does not try to locate the source of necessity among contingent truths. He advocates what he calls “non-transmissive” explanation on modality. One example of a transmissive explanation of modality is mentioned earlier: $\Box(3^2 = 9 \text{ or snow is white})$ because $\Box 3^2 = 9$ and $\Box p \rightarrow \Box(p \vee q)$. Here not only the truth of $3^2 = 9$ but also the necessity of it is used for the explanation, i.e., the explanation points out that the necessity is transmitted from the explanans to the explanandum. In general, a transmissive explanation will take the form of $\Box p$ because $\Box q$, where q itself cannot explain $\Box p$. Perhaps Blackburn is correct if this is the only available form of explanation. It may help us to sort out some “fundamental” necessities from non-fundamental ones, but does not give us the general source of necessity, for it is precisely the fundamental necessities that needs to be accounted their sources.

Non-transmissive explanations, on the other hand, take the form of $\Box p$ because q , where q is a proposition that is necessarily true “but in which what explains the necessity of the explanandum is not q ’s *necessity*, but its *truth* simpliciter.” (Hale 2002, p.311) Contra Blackburn, Hale explicitly takes some necessarily truths to be the general source of necessity. Since q is also a necessary truth, the necessity of $\Box q$ would also be explained via the truth of a proposition which is necessarily true, say r . But that does not imply that the necessity of p is better explained by r . For, it is the truth simpliciter of q that explains the necessity of p , not its necessity. Rather, if the truth of q is explained ultimately by the truth of s , it would make sense to say the necessity of p is explained ultimately by s .

Therefore, if we succeed in giving a non-transmissive explanation to all, or at least the fundamental necessities, it will suffice for being an account for the general source of necessity, which does not appeal to further necessities. This is usually done by specifying

what kinds of necessary truths explains the related necessities. Hale’s preferred view is that “Necessarily p because $\Box_{X_1, \dots, X_n} p$ ” (Hale 2013, p.152): the necessity of any proposition p is explained by it being true in virtue of, or being essentially true of some objects $X_1, \dots, X_n$. But this is not the sole possible explanation. For instance, modal realist would assert that the necessity of any proposition p is explained by p being true at all possible worlds. Now, if the notion of possible worlds can be specified without modal notions, this is a genuine case of a non-transmissive explanation: it is necessary that p is true at all possible worlds (given that p is a necessary truth), but the truth of p being true at all possible worlds, not its necessity, does the explanation work.

It is rational to expect any sensible essentialist account on modality would be non-transmissive. Essentialists typically accept the biconditional $\Box A \leftrightarrow \exists F \Box_F A$ (Fine 2020, p.461): a sentence or a proposition is necessary if and only if it is true in virtue of the nature of some objects or properties. And if a essentialist particularly holds that the necessity of A is explainable by it being true in virtue of the nature of some *objects*,¹ it is a theorem under the standard axioms of the logic of essence that $\Box_F A \rightarrow \Box_F \Box_F A$, where the predicate F is rigid, i.e., denotes a property of being identical to some object (Fine 1995a, p.255, see THEOREM 6 (ii)). That is, if A is true in virtue of the nature of F , the fact that A is true in virtue of the nature of F is true in virtue of the nature of F . It implies that, since essentialists accept the biconditional above, $\Box_F A$, the explanan of the necessity of A , must also be necessary. But since essentialists would want $\Box_F A$, not the necessity of $\Box_F A$ to explain the necessity of A , it constructs the exact case of the expalnation of necessity where the explanans are necessarily true yet its *truth* simpliciter, not necessity explaining the necessity of the explanandum.

Note that there can be slightly different view on how essence can account for the source of necessity among essentialists. For Kit Fine, being metaphysical necessary is *defined* to being true by the the essence of all objects (Fine 1994, 1995a). The notion of metaphysical necessity is therefore reduced in terms of essence. This implies that Fine would agree that the necessity of p would be grounded on p being true in virtue of the nature of all objects. Following the notation presented in Fine (2015), such a definition which gives necessary and sufficient condition of being necessary yields [Given x where sentence(x): $\Box_{\forall} x \leftrightarrow_x \Box x$], where $\forall$ is defined as $\lambda x (\Pi x (x = x))$. That means that something being true in virtue of the nature of all objects is both (partial) ground and essence of something being necessary. Hale, on the contrary, explicitly claims his account to be non-reductive. Although he agrees with Fine that metaphysical necessity is grounded in the nature of things, he also takes modality as “fundamental and irreducible” (Hale 2013, Ch.3). Lowe (2008) also suggests that all metaphysical necessity is grounded on essence, where essence just is the nature of an object, not a further entity. It is not very clear whether Lowe would agree that necessity can be defined in terms of essence, but it seems likely that Lowe would not equate metapyhsical necessity and the maximal essence as Fine does, for it is doubtful whether Lowe would include mediate and consequential essence (Fine 1995c) when he speaks of essence.

One restriction for all essentialists is that the notion of essence should not be understood

¹Fine (1994) writes: “The metaphysically necessary truths can then be identified with the propositions which are true in virtue of the nature of all *objects* whatever” (p.9, italics added); Hale (2013) also writes: “Metaphysical necessities might plausibly be identified as those which hold in virtue of the nature of *things* in general.” (p.150, italics added). Additionally, the variable X , although not formalized, are almost always introduced explicitly as variable ranging over various kinds of entities.

in terms of modal notions. For example, if the definition of Fa being true in virtue of the nature of a is just that a necessarily having F , then the source of necessity of Fa cannot be that Fa being true in virtue of the nature of a , for that would still leave *de re* necessity unexplained, or ending up being circular. It is parallel to the reason why possible world theorists should not define possible worlds as worlds (mereological aggregates) that are metaphysically possible. But if essence should be understood independently from necessity, how can one ensure that all essential facts are necessary?

It seems that essentialists must accept that all essential truths, what is true in virtue of the nature of something are necessary. To acknowledge contingent essence, one must give a criteria demarcating them with necessary essence without appealing to the notion of essence. That is, one must identify the non-essentialist source of the necessity of necessary essences, which contingent essences lack. And this apparently undermines essentialism.

Some skeptics question exactly that whether all essential truths are necessary. To put it formally, essentialists are committed into what Mackie (2020) calls “Necessity Principle (NP)”:

$$(NP) \quad \Box_F A \rightarrow \Box A^2$$

where A is any rigid predicate and A is any sentence. But what ensures (NP)? One common understanding of essence among essentialists is by *real definition*: roughly, the essence, or the identity of an object is captured or given by its real definition, and therefore what is true in virtue of the nature of something is what is true by the definition of something (Fine 1994, p.2; Hale 2013, p.132; Lowe 2012, p.919). Alternatively, Correia and Skiles (2017) suggest to understand essence in terms of generalized identity, expressed by the identity statements such as “for a thing to be a bachelor is for it to be an unmarried adult male” or “for the Atlantic Ocean to be filled with water is for it to be filled with H_2O molecules” (p.643). Here being F expresses the full essence of being G if and only if to be G is to be F ; it expresses partial essence of being G if and only if there is some H such that to be G is to be both F and H (pp.649-50). The skeptic charge is that there seems no direct evidence that the nature of real definition or generalized identity can ground (NP) alone. How can essentialists “deliver a modal rabbit out of a non-modal hat” (Mackie 2020, p.252)?

In the case of definitional essentialists, Hale tried to give an argument. Suppose $\Phi\alpha$ is true in virtue of the identity of α . Now suppose it is possible that $\neg\Phi\alpha$, which follows that it is possible that $\alpha = \beta \wedge \neg\Phi\beta$. But β cannot be identical to α for it does not satisfy the condition to be identical to α , which is Φ . Therefore $\neg\alpha = \beta$, contradiction. Since contradictions are impossible, we can conclude that if $\Phi\alpha$ is true in virtue of the identity of α , $\Phi\alpha$ is necessary (Hale 2013, p.133).

But it has been pointed out that the argument does not successfully answer to the skepticism toward (NP) (Romero 2019, pp.125-7; Leech 2021, pp.893-5). For the proposed argument to succeed, it must be assumed that Φ is *necessarily* the condition for some arbitrary objects to be identical to α . Either $\Box(\Box_\alpha\Phi)$ or $\Box\Phi\alpha$ is required. But what is assumed is only that Φ is *actually* the condition of the identity. To hold that the assumption implies $\Box\Phi\alpha$ simply begs the question. And while $\Box_\forall(\Box_{\lambda x.x=\alpha}\Phi)$ is provable from $\Box_{\lambda x.x=\alpha}\Phi$, the inference by Finean definition from $\Box_\forall A$ to $\Box A$ again begs the ques-

²This formalization follows Fine (2020).

tion, for that is just another form of (NP). Moreover, Hale himself would not agree to the identification between necessary truths and the proposition true in virtue of the nature of all objects.

In fact, one can construct a proof for (NP) once she accepts the Finean definition of necessity. It is an axiom in the logic of essence that $F \subseteq G \rightarrow (\Box_F A \rightarrow \Box_G A)$, where “ $R \subseteq G$ ” denotes $\Pi x(Rx \rightarrow Gx)$ (Fine 1995a, pp.246-7). Since for any rigid predicate F $F \subseteq \bigvee$, it follows that $\Box_F A \rightarrow \Box_{\bigvee} A$. And since the necessity operator $\Box$ is just defined as abbreviating $\Box_{\bigvee}$, we can derive $\Box_F A \rightarrow \Box A$, which is (NP) itself. But there is another reason why skeptic regard the definition to be unsatisfactory (Romero 2019, p.129-36). As it was mentioned, Fine defines necessary truths as propositions that are true by the essence of all objects. He writes $\Box_{\bigvee} S$ for S being essentially true in virtue of all objects, where $\bigvee$ is defined as $\lambda x(\Pi x(x = x))$, and Π is a quantifier ranging over “all metaphysically possible objects.” (Fine 1995a, p.250). Persuasively, since the problem of which objects are possible depends on the notion of modality, to conceive necessary truths as maximal essence, the essence of all *possible* objects, is circular. In the same paper Fine also writes “[...] one might operate with a broader conception of possible object than is customary in modal logic” (ibid, p.244), but it is still unclear what the “broder conception of possible object” is.

To understand essence in terms of generalized identity may seem to have better chance for one to advocate (NP). “As with objectual identity, generalized identity is tightly linked to metaphysical necessity” (Correia and Skiles 2017, p.656). Following Correia and Skiles, let “ $p \equiv q$ ” denote the generalized identity between two predicates or propositions p and q . As it is the case of the objectual identity, $p \equiv p$ is a theorem³ and therefore in modal logic it follows that $\Box(p \equiv p)$. And given $p \equiv (q \wedge r)$ [$p \equiv q$], which means that q is a partial [full] essence of p , by substitutability we get $\Box(p \equiv (q \wedge r))$ [$\Box(p \equiv q)$]. That is, assuming q is a partial [full] essence of p , it follows that *necessarily* q is a partial [full] essence of p .

This actually proves something stronger than (NP): namely, $\Box_F A \rightarrow \Box \Box_F A$. However, since if something is true in virtue of the nature of something it would also be true *simpliciter*, it would be reasonable to think that implies (NP). The problem is that the proof presupposes $\Box(p \equiv p)$; that the generalized identities are necessary. How can essentialists explain that particular necessity? It was argued that the necessity of $\Box(p \equiv p)$ cannot be explained by the essence of generalized identity (Leech 2021, pp.901-2; cf. Romero 2019, p.138). For example, suppose that the necessity is explained by that for two things to be generically identical is partially to bear a reflexive relation. Grant, but what explains the necessity of the partial generalized identity between generalized identity and reflexive relation? Here it seems we cannot appeal to the necessity of $p \equiv p$, for that exactly is what it is explaining, not something that is explained by. Therefore, to appeal to the essence of generalized identity itself cannot serve as the final source of the necessity of all generalized identity.

Leech seems to conclude that, in explaining the necessity of the generalized identity that expresses the (partial) essence of generalized identity itself “*one should not appeal to the essence of anything to account for this necessity*”. She further considers this constraint as “the crucial and general objection” on the necessity of essence (p. 902).

³For they take generalized identity to be reflexive (p.645)

I disagree. I think Leech overlooks one remaining possibility of appealing to the essence: that generalized identities, including $p \equiv p$ are necessary because of the essence of *being necessary*. In my view, this also may apply directly on (NP). Every propositions true in virtue of the nature of something or all things are also necessarily true, not because of the nature of essence but because of the nature of what is it to be necessary. In section 2 and 3, I will argue that essentialists must accept that (NP) is true in virtue of the nature of what is it to be necessary.

2 Non-transmissive Explanation and the Essence of Being Necessary

I argued that any sensible essentialist account on necessity would hold that essence provides a non-transmissive explanation on necessity. Non-transmissive explanations, as mentioned, take the form of $\Box p$ because q , where q is also necessarily true. Usually, the role of q will be taken by a proposition *about* p . For example, that *p is true in all possible worlds*, or that *p is true in virtue of the nature of something*, and so on. Let S denote some status, or criteria that a proposition can satisfy, for example being true in all possible worlds. Then we can modify the form of non-transmissive explanation into $\Box p$ because $S(p)$, where p necessarily satisfies S . A theory on the general source of necessity will then give an account of what the status S is. Essentialists are those who hold that the status S is *being true in virtue of the nature of something*.

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